Vineeth Appala

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SUMMARY

Data Science & ML Engineer with 6+ years of experience in delivering end-to-end AI/ML solutions. Proven track record in building scalable data pipelines, developing high-performing machine learning and deep learning models (NLP, CV, recommender systems), and deploying solutions using modern MLOps practices. Expert in Python, SQL, Spark, TensorFlow, and cloud platforms (AWS, GCP). Adept at visual storytelling with Tableau and Power BI, with strong collaboration and stakeholder engagement skills.

SKILLS

- **Machine Learning:** XGBoost, SVM, Random Forest, Logistic Regression, Naive Bayes, KNN, Bayesian HMM
- **Frameworks & Libraries:** TensorFlow, Keras, Caffe, Matplotlib, Seaborn, Numpy, Pandas, Sci-kit Learn, OpenCV
- **Programming Languages:** Python, R, SQL
- **Model Development:** Neural Networks (CNN, RNN), Feature Engineering, EDA, Model Validation, Sentiment Analysis, Text Classification, Named Entity Recognition, Tokenization, Text Mining
- **Data Engineering:** Data Wrangling, ETL Pipelines
- **Big Data Tools:** Hadoop, HDFS, MongoDB, Teradata
- **Cloud & MLOps:** GCP (BigQuery, Vertex AI, Cloud Storage, Data Studio), AWS (Redshift, SageMaker, S3), Databricks, Airflow
- **Visualization & Deployment:** Power BI, OLAP Cubes, Dashboards, Tableau, Excel
- **Version Control & CI/CD:** Git, Jira, Confluence

EXPERIENCE

Mellowsoft
Data Science & ML Engineer — **Oct 2024 - Current**

- Engineered a sophisticated RAG model with vector database integration to retrieve pertinent information from a 10TB+ document corpus, enhancing contextual relevance of model-generated responses by 25% and improving user satisfaction scores.
- Implemented a document retrieval mechanism using dense retrievers, pre-trained Sentence-BERT embeddings, and Faiss indexing, achieving 90% retrieval accuracy for relevant context from a structured knowledge graph-based database.
- Integrated the retrieval system with an LLM (T5/GPT-3) leveraging attention mechanisms and fine-tuning, generating context-aware answers that improved business KPI alignment by 15% for data-driven insights.
- Preprocessed and cleansed large datasets from diverse sources like customer service logs, technical manuals, and FAQs using data orchestration pipelines and NLP tokenization, ensuring 95% data quality for retrieval model inputs.
- Evaluated model performance using precision, recall, BLEU, and ROUGE scores, achieving a 0.85 F1 score and implementing A/B testing to ensure robustness and reliability in generated outputs.
- Optimized the model for scalability by tuning hyperparameters, implementing efficient caching with Redis, and leveraging distributed computing, reducing retrieval latency by 30% and boosting generative accuracy.
- Deployed the RAG-based conversational AI model to production via a RESTful web API on Kubernetes, utilizing MLOps with model monitoring and drift detection to achieve 99.9% uptime and support continuous iterative improvements.

Toyota
Data Scientist — **Dec 2021 - Sep 2024**

- Developed and deployed a Collaborative Filtering recommendation model with hyperparameter optimization, boosting website user engagement by 20% through natural language processing (NLP)-enhanced analysis of 1M+ monthly clickstream records, validated via A/B testing frameworks.
- Processed multi-terabyte datasets in BigQuery (GCP) with Apache Spark, optimizing data lake-integrated SQL pipelines to reduce query runtime by 30% and enable real-time inventory forecasting.
- Engineered 50+ features using TF-IDF, one-hot encoding, feature scaling, and dimensionality reduction, improving gradient boosting-based recommendation model accuracy by 12% for personalized dealer orders.
- Created interactive dashboards in Data Studio, Tableau, and Looker, visualizing supply chain KPIs and user behavior with data governance protocols, accelerating stakeholder decisions by 25%.
- Leveraged Vertex AI (GCP) to build XGBoost, Random Forest, and neural network models, predicting webpage navigation with 85% accuracy using ensemble learning, increasing click-through rates by 10%.
- Automated preprocessing pipelines with Python (pandas, NumPy), Apache Airflow, and Kubernetes, cutting manual processing time by 40% for globally distributed supply chain data.
- Collaborated cross-functionally to align AI-driven solutions and predictive analytics with business objectives, reducing excess inventory by 10% through time-series forecasting.
- Implemented MLOps practices using Git, Docker, Jenkins CI/CD, and model monitoring, ensuring 99% uptime for scalable recommendation systems.

Watergrass
Data Scientist — **Jan 2020 - May 2021**

- Analyzed 40+ years of historical data from 30+ organizations, applying descriptive statistics, outlier treatment, imputation, and data wrangling, ensuring robust datasets with data quality frameworks for 500K+ records.
- Engineered 20+ predictive features, utilizing standardization, normalization, and feature selection to enhance logistic regression model performance by 10% for donor segmentation.
- Designed dynamic Power BI dashboards with DAX and real-time APIs, tracking campaign KPIs and donor trends under data governance standards, reducing reporting time by 35%.
- Built and deployed machine learning models using Python (scikit-learn), R, and LightGBM, achieving 80% accuracy in predicting high-value donors with cross-validation, driving a 12% increase in funds raised.
- Streamlined data pipelines with SQL, ETL processes, and Apache NiFi, automating CRM-integrated data ingestion to cut preprocessing time by 25%.
- Prototyped solutions with machine learning teams, integrating RESTful APIs and real-time streaming for donor insights, improving campaign targeting efficiency by 18%.
- Translated complex analytics into strategic recommendations using statistical modeling and business intelligence, presenting to C-level stakeholders to secure 10% more funding.

- Utilized statistical tools like SPSS, Excel, and Python with Bayesian analysis to validate trends, ensuring scalable and reproducible predictive models.

Nuronics Labs
Data Scientist **Jan 2018 - May 2019**

- Processed 1,000 patient datasets, each with ~150 DICOM files, using pydicom for metadata extraction and data augmentation, managing 10TB+ of imaging data with data lake architecture.
- Applied image enhancement techniques with OpenCV, SciPy, and histogram equalization, reducing noise by 15% and improving contrast for semantic segmentation of lung tissue.
- Developed a Convolutional Neural Network (CNN) in Python (TensorFlow, Keras) with transfer learning, optimizing hyperparameters to achieve 76% testing accuracy for cancer classification.
- Automated preprocessing pipelines with Python scripts and Apache Airflow, cutting manual image preparation time by 30% and enabling scalable batch processing.
- Collaborated with product teams to integrate CNN models into clinical workflows using ONNX, improving clinician decision-making speed by 25% with explainable AI.
- Validated model performance using ROC-AUC, confusion matrices, and precision-recall curves, achieving a 0.82 AUC score with model interpretability for reliable predictions.
- Contributed to MLOps workflows, deploying models via AWS SageMaker, Docker containers, and Kubernetes orchestration, maintaining 98% system uptime.
- Presented findings to healthcare stakeholders, translating technical results into actionable clinical insights using SHAP values, securing funding for further AI diagnostics research.

EDUCATION

Master of Science, Business Analytics **Aug 2019 - Sep 2021**
University of New Hampshire
Bachelor of Technology, Electronics and Communication Engineering **Aug 2015 - May 2019**
Manipal University Jaipur

CERTIFICATIONS

- **AWS Certified Machine Learning – Specialty**
- **Google Professional Data Engineer**
- **Google Professional Machine Learning Engineer**
- **Generative AI with Large Language Models Certificate – DeepLearning.AI**